Pump It Up: Practice Test 1

Time: 50 minutes. Work with your partner at 13 stations, about 3 1/2 minutes each. When the coach calls 'Rotate!', move to the next station; you may not skip ahead or go back, even if a station is empty. Bring only pencils: no notes, flashcards, or books, just like the tournament.

Partner names: _____ and _____ Score: _____

Station 1: Heart map

1. Look at the floor heart diagram. Name the parts at labels 1, 2, 3, and 4.

2. Name the parts at labels 5, 6, 7, and 8.

3. Blood goes from label 2 into label 1. Which valve (with no label) does it pass through?

4. Which 2 numbered chambers hold oxygen-rich blood?

Station 2: Trip through the heart

5. Blood coming back from the body enters which chamber first?

6. Put these in the order blood visits them, starting from the body: left ventricle, lungs, right atrium, left atrium, right ventricle, aorta.

7. Which valve is between the left atrium and the left ventricle?

◯ A Tricuspid ◯ B Mitral ◯ C Aortic ◯ D Pulmonary

8. Blood leaves the right ventricle. Which valve does it pass through, and where is it going?

Station 3: Rooms and doors

9. What is the job of the atria? What is the job of the ventricles?

10. The 'lub-dub' sound of a heartbeat comes from what?

11. A valve does not close all the way and leaks. What goes wrong?

12. Which side of the heart pumps blood to the lungs?

◯ A The right side ◯ B The left side

Station 4: What's in blood?

13. The yellowish liquid part of blood that carries food, water, and wastes is called _____.

14. About how much of your blood is plasma?

◯ A About 1 tenth ◯ B About half ◯ C Almost all of it

15. Jake has an infection. Which blood cells fight the germs?

16. You scrape your knee. Which part of blood helps make a clot to stop the bleeding?

Station 5: Blood highways

17. Which vessels have thick, stretchy walls? Why do they need them?

18. Which vessels have walls only one cell thick, so oxygen can pass through?

19. Veins have little valves inside. Why?

20. When you feel your pulse at your wrist, which kind of vessel are you feeling?

Station 6: Airway

21. Put these in order from where air enters to where it ends: bronchioles, larynx, trachea, nasal cavity, alveoli, pharynx, bronchi.

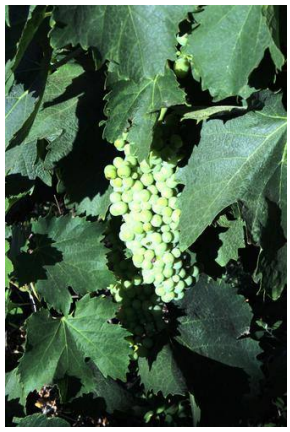
22. Which part is the voice box?

23. Which part has stiff rings that keep it open like a vacuum hose?

24. Why is breathing in through your nose better than through your mouth?

Station 7: Grape model

A scientist uses this bunch of grapes as a model of part of the lungs.



25. In this model, what do the grapes stand for? What do the stems stand for?

26. What tiny blood vessels wrap around each air sac?

27. Why is it helpful to have millions of tiny air sacs instead of one big bag?

28. How many lobes does the right lung have? The left lung?

Station 8: Breathing muscle

29. Look at the balloon lung model. Pull the bottom balloon down, then push it up. What happens to the small balloon each time?

30. When you breathe in, the diaphragm tightens and moves _____, and air rushes _____.

31. When you breathe out, what does the diaphragm do?

Station 9: Team work

32. In the alveoli, which gas moves into the blood? Which gas moves out of the blood?

33. Your muscle cells use oxygen. What waste gas do they make?

34. Blood going from the heart to the lungs is oxygen-_____. Blood coming back from the lungs to the heart is oxygen-_____.

35. In 2 sentences, explain how the heart and lungs work as a team.

Station 10: Jump rope data

Jada's data: At rest, pulse 80 beats per minute, breathing 18 breaths per minute. Right after 2 minutes of jump rope: pulse 140, breathing 40. After 5 minutes of rest: pulse 95, breathing 22.

36. What happened to her pulse and breathing right after jump rope? Why?

37. Why did both numbers go back down after she rested?

38. Jada counts 30 heartbeats in 15 seconds. What is her pulse in beats per minute? Show your math.

Station 11: Heart and blood case files

For each person, name the condition and the main body part or system it affects.

39. Tomas was tested as a baby. His red blood cells are shaped like a C, get stuck in small vessels, and cause pain.

40. Ms. Lee suddenly cannot lift her right arm, one side of her face droops, and her words are slurred.

41. Mr. Brown is shoveling snow and feels crushing chest pain because an artery to his heart is blocked.

42. Grandma's heart pumps weaker than it should. Her ankles swell and she gets out of breath on stairs.

Station 12: Lung case files

For each person, name the condition and one treatment or way to prevent it.

43. When pollen is high, Ava coughs, wheezes, and her airways get narrow.

44. Ben got a gene from each parent that makes thick, sticky mucus clog his lungs. He wears a shaking vest every day.

45. Mr. Ruiz smoked for 35 years. His airways and air sacs are damaged, and it is hard to breathe all the way out.

46. Ms. Hart never smoked, but her basement had high radon gas. A scan found a lump of cells growing out of control in her lung.

Station 13: Causes and healthy habits

47. Which 2 conditions on our list are genetic (passed from parents)?

48. Can you catch cystic fibrosis or asthma from a friend? Explain.

49. What is the number one habit that causes COPD and lung cancer?

50. Name 2 healthy habits that lower the risk of heart attack and stroke.

Station 8: Breathing muscle Coach: set out one balloon lung model from Lesson 3.	
29	Pull down: the small balloon fills with air. Push up: it lets the air out.
30	Down; in.
31	It relaxes and moves up, pushing air out of the lungs.
Station 9: Team work	
32	Oxygen moves into the blood. Carbon dioxide moves out (and is breathed out).
33	Carbon dioxide.
34	Poor; rich.
35	The lungs bring oxygen into the blood and remove carbon dioxide. The heart pumps the oxygen-rich blood to the body and sends the oxygen-poor blood back to the lungs.
Station 10: Jump rope data	
36	Both went up. Her working muscles needed more oxygen and made more carbon dioxide, so her heart pumped faster and she breathed faster.
37	Her resting muscles needed less oxygen, so her heart and lungs could slow down.
38	30 x 4 = 120 beats per minute.
Station 11: Heart and blood case files	
39	Sickle cell anemia. Blood (red blood cells).
40	Stroke. The brain (a blood vessel to the brain is blocked or leaking). Call 911.
41	Heart attack. The heart muscle. Call 911.
42	Congestive heart failure. The heart (fluid backs up into the legs and lungs).
Station 12: Lung case files	
43	Asthma. Treatment: rescue inhaler or controller medicine; avoid triggers like pollen.
44	Cystic fibrosis. Treatment: daily airway clearing (vest), inhaled medicine, enzyme pills, newer medicines.
45	COPD. Prevention: do not smoke. Treatment: quit smoking, inhalers, extra oxygen.
46	Lung cancer. Prevention: test the home for radon, avoid smoke. Treatment: surgery, chemotherapy, radiation.
Station 13: Causes and healthy habits	
47	Sickle cell anemia and cystic fibrosis.
48	No. They are not caused by germs, so they are not catching. CF is genetic, and asthma runs in families and is set off by triggers.
49	Smoking (and breathing other people's smoke).
50	Any 2: exercise, eat healthy food with less salt and fat, do not smoke, get checkups to keep blood pressure healthy.